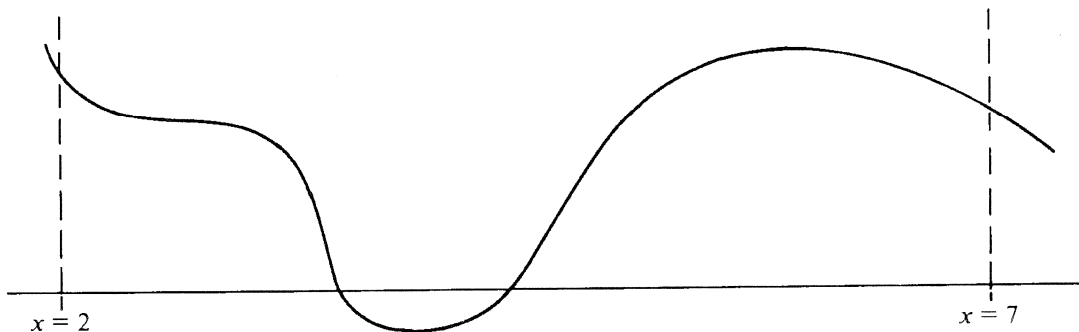
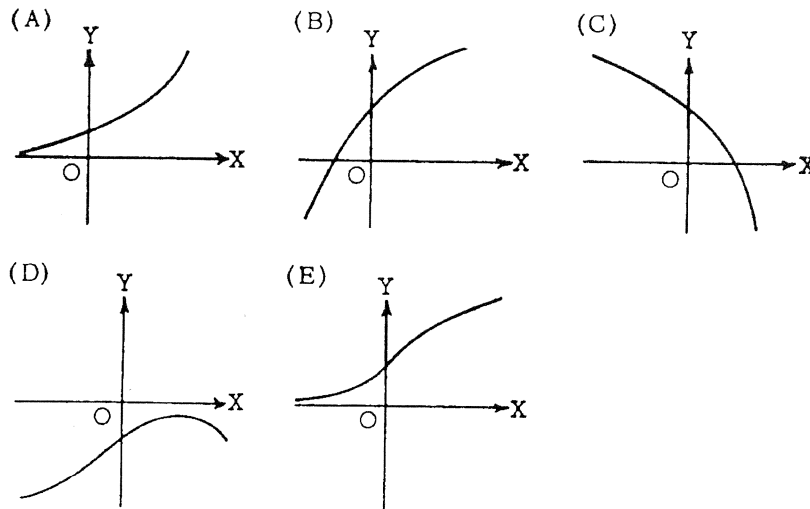


- BC 16. If y is a function x such that $y' > 0$ for all x and $y'' < 0$ for all x , which of the following could be part of the graph of $y = f(x)$?

1969

NC



- BC 20. The graph of $y = f(x)$ on the closed interval $[2, 7]$ is shown above. How many points of inflection does this graph have on this interval?

1985

NC

- (A) One (B) Two (C) Three (D) Four (E) Five

4. Let h be a function defined for all $x \neq 0$ such that $h(4) = -3$ and the derivative of h is given by

2001
BC NC $h'(x) = \frac{x^2 - 2}{x}$ for all $x \neq 0$.

- (a) Find all values of x for which the graph of h has a horizontal tangent, and determine whether h has a local maximum, a local minimum, or neither at each of these values. Justify your answers.
 (b) On what intervals, if any, is the graph of h concave up? Justify your answer.
 (c) Write an equation for the line tangent to the graph of h at $x = 4$.
 (d) Does the line tangent to the graph of h at $x = 4$ lie above or below the graph of h for $x > 4$? Why?